Supplementary Material: Structural Rank, Decision Entropy, and Thermodynamic Selection in Finite Information-Processing Systems

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1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Lean Handle / Source	ID	Lean Handle / Source
BA1	Physics.BoundedAcquisition.BoundedRegion paper4/DecisionQuotient/Physics/BoundedAcquisition.lean	BA2	Physics.BoundedAcquisition.acquisition_rate_bound paper4/DecisionQuotient/Physics/BoundedAcquisition.lean
BA3	Physics.BoundedAcquisition.acquisitions_are_transitions paper4/DecisionQuotient/Physics/BoundedAcquisition.lean	BA4	Physics.BoundedAcquisition.one_bit_per_transition paper4/DecisionQuotient/Physics/BoundedAcquisition.lean
BA5	Physics.BoundedAcquisition.resolution_reads_sufficient paper4/DecisionQuotient/Physics/BoundedAcquisition.lean	BA6	Physics.BoundedAcquisition.srank_le_resolution_bits paper4/DecisionQuotient/Physics/BoundedAcquisition.lean
BA7	Physics.BoundedAcquisition.energy_ge_srank_cost paper4/DecisionQuotient/Physics/BoundedAcquisition.lean	BA8	Physics.BoundedAcquisition.srank_one_energy_minimum paper4/DecisionQuotient/Physics/BoundedAcquisition.lean
BA10	Physics.BoundedAcquisition.counting_gap_theorem paper4/DecisionQuotient/Physics/BoundedAcquisition.lean	EI1	ThermodynamicLift.energy_ge_kbt_nat_entropy paper4/DecisionQuotient/ThermodynamicLift.lean
IT3	DecisionQuotient.quotientEntropy_le_srank_binary paper4/DecisionQuotient/Information.lean	IT4	DecisionQuotient.numOptClasses_le_pow_srank_binary paper4/DecisionQuotient/Information.lean
L17	Leverage.compose_dof Leverage/Foundations.lean	L19	Leverage.composition_dof_additive Leverage/Theorems.lean
L43	dof_eq_srank Leverage/BridgeToDQ.lean	L44	dof_one_iff_max_leverage Leverage/Foundations.lean
L45	england_replication_inequality Leverage/BridgeToDQ.lean	L46	incoherent_srank_gt_one Leverage/BridgeToDQ.lean
L47	max_coherence_forces_tractability Leverage/BridgeToDQ.lean	L49	srank_energy_lower_bound Leverage/BridgeToDQ.lean
L51	ssot_srank_one Leverage/BridgeToDQ.lean	L52	succ_le_two_pow Leverage/BridgeToDQ.lean
L53	sufficiency_conp_hard	L54	thermodynamic_selection Leverage/BridgeToDQ.lean
L55	thermodynamic_selection_unconditional Leverage/BridgeToDQ.lean	L56	tractable_bounded_core paper4/DecisionQuotient/ClaimClosure.lean

(continued...)

ID	Lean Handle / Source	ID	Lean Handle / Source
ORA1	oracle_arbitrary paper2/Ssot/Coherence.lean	PH14	PhysicalComplexity.p_eq_np_physically_ impossible_of_collapse_map paper4/DecisionQuotient/Physics/ PhysicalHardness.lean
PH15	PhysicalComplexity.p_eq_np_physically_ impossible_canonical paper4/DecisionQuotient/Physics/ PhysicalHardness.lean	PH26	PhysicalComplexity.no_collapse_of_bounded_ budget_pos_cost_exp_lb paper4/DecisionQuotient/Physics/ PhysicalHardness.lean
QT1	DecisionProblem.quotient_is_coarsest paper4/DecisionQuotient/Quotient.lean	QT2	DecisionProblem.quotientMap_preservesOpt paper4/DecisionQuotient/Quotient.lean
QT3	DecisionProblem.quotient_represents_opt_equiv paper4/DecisionQuotient/Quotient.lean	QT7	DecisionProblem.quotient_has_unique_ factorization paper4/DecisionQuotient/Quotient.lean
W1	Physics.single_future_zero_cost paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean	W2	Physics.transportCost_pos_of_offDiag paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean
W3	Physics.integrity_is_centroid paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean	W4	Physics.wasserstein_bridge paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean

2 Claim-to-Lean Handle Mapping

This section maps each paper claim to its corresponding Lean formalization.

Paper claim	Lean handle
Corollary 4.4: Unique Minimum-Cost Regime	BA8, L54, L55
Corollary 5.9: $P = NP$ No-Go Transfer	PH26, PH15, PH14
Corollary 3.3: Higher-Rank Regime	L46, L51
Corollary 3.2: Rank-One Regime	L51
Proposition 2.5: Bounded Region	BA1
Definition 2.3: Bounded Decision System	L17, L19
Definition 2.13: Canonical Decision Problem	QT2, QT7, QT1, QT3
Theorem 2.6: Bounded Acquisition Rate	BA1, BA2
Theorem 5.1: Coherent Single-Source Regime	ORA1
Theorem 2.4: Counting Gap	BA10
Theorem 2.7: Discrete Acquisition	BA3
Theorem 3.1: DOF–Structural-Rank Identity	L43
Theorem 4.3: Energy–Information Duality	IT3, EI1, L43
Theorem 4.1: Rank Controls Exact-Resolution Cost	BA7, BA6, L43
Theorem 5.6: England Replication Inequality	L45
Theorem 3.6: Decision-Entropy Bound	IT3, L43
Theorem 5.7: Finite-Budget No-Collapse	PH26
Theorem 5.5: Convergence	L43, L44, L47, L55
Theorem 3.4: Minimum Physical Bit Operations	BA5, BA6, L43, L46
Theorem 3.5: Decision-Class Bound	IT4, L43
Theorem 2.8: One Transition, One Bit	BA3, BA4

Paper claim	Lean handle
Theorem 5.2: Rank Identification	L43, L46, L51
Theorem 4.2: Rank-One Ground State	BA8, L54, L55
Theorem 2.9: Resolution Requires a Sufficient Coordinate Set	BA5
Theorem 5.4: Thermodynamic Selection	BA8, L49, L54, L55
Theorem 5.3: Tractable Sufficiency at Rank One	L47, L53, L56

Auto summary: mapped 26/26 (full=26, derived=0, unmapped=0).

3 Proof Hardness Index

Paper handle	Hardness profile	Regime tags	Lean support
cor: minimum-cost-regime	unspecified	-	BA8, L54, L55
cor:pnp-nogo	unspecified	-	PH26, PH15, PH14
cor:rank-above-one	unspecified	-	L46, L51
cor:rank-one	unspecified	-	L51
prop:bounded-region	unspecified	-	BA1
prop:dof-additive	unspecified	-	L17, L19
prop: optimizer-quotient	unspecified	-	QT2, QT7, QT1, QT3
thm: bounded-acquisition	unspecified	-	BA1, BA2
thm: coherent-single-source	unspecified	-	ORA1
thm:counting-gap	unspecified	-	BA10
thm: discrete-acquisition	unspecified	-	BA3
thm:dof-srank	unspecified	-	L43
thm:energy-entropy	unspecified	-	IT3, EI1, L43
thm:energy-rank	unspecified	-	BA7, BA6, L43
thm:england	unspecified	-	L45
thm:entropy-bound	unspecified	-	IT3, L43
thm: finite-budget-no-collapse	unspecified	-	PH26
thm:five-way	unspecified	-	L43, L44, L47, L55
thm: min-bit-operations	unspecified	-	BA5, BA6, L43, L46
thm:numopt-bound	unspecified	-	IT4, L43
thm: one-transition-one-bit	unspecified	-	BA3, BA4
thm: rank-identification	unspecified	-	L43, L46, L51
thm:rank-one-ground	unspecified	-	BA8, L54, L55
thm: resolution-sufficient	unspecified	-	BA5
thm: thermodynamic-selection	unspecified	-	BA8, L49, L54, L55
thm: tractable-rank-one	unspecified	-	L47, L53, L56

Auto summary: indexed 26 claims by hardness profile (unspecified=26).

4 Assumption Ledger

4.1 Assumption Ledger (Auto)

Source files.

- `Leverage/BridgeToDQ.lean`
- `Leverage/FiveWayEquivalence.lean`
- `Leverage/Physical.lean`

Assumption bundles and fields.

- (No assumption bundles parsed.)

Conditional closure handles.

- (No conditional theorem handles parsed.)

First-principles Bayes derivation handles.

- (No Bayes-derivation theorem handles found.)

Compact Bayes handle IDs.

- (No compact Bayes alias IDs found.)